

Jagraj Gill

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EDUCATION

Simon Fraser University,

December 2027

Bachelor of Science in Computing Science

Relevant Coursework: Data Structures & Algorithms I, Data Structures & Algorithms II, Software Engineering, Databases

EXPERIENCE

Software Engineer Intern, Savi Finance – Toronto, ON

May 2025 – December 2025

- Developed an **AI-powered task scheduling system** that ranked tasks by priority, reducing planning time by **10%**
- Engineered shared modules across **React** and **React Native** backed by **MongoDB** services for a fintech SaaS platform serving **1000+ users**, reducing release cycles by **15%**
- Built and integrated **image upload workflow** using **AWS S3** and **GraphQL**, supporting custom image previews and improving upload reliability while reducing image load failures by **10%**
- Refactored **login UI and validation** using **React** and **custom hooks**, reducing user login errors by **45%**

Hackathon, Storm Hacks 2024 – Burnaby, BC

May 2024

- Designed and deployed **REST APIs** with **Node.js** and **Express.js** to enable real-time data updates, **reducing system response time by 30%**
- **Directed a team** during a 24-hour hackathon, coordinating tasks and project delivery under strict deadlines

PROJECTS

IoT Climate Monitoring Dashboard | ESP-IDF, FastAPI, SQLite, JavaScript, HTML, CSS

May 2026

- Built an **end-to-end IoT climate monitoring system** integrating microcontroller firmware, I2C sensor communication, OLED display output, REST APIs, SQLite storage, and a live web dashboard
- Developed **ESP-IDF firmware** to collect temperature and humidity readings, update an OLED interface, and transmit **structured JSON** data to a FastAPI backend over Wi-Fi
- Engineered **24-hour climate tracking** with SQLite and REST endpoints, enabling a JavaScript dashboard to display live readings, device status, and trend visualizations

AI Study Pal | Python, FastAPI, Gemini API, HTML, CSS, JavaScript

April 2026

- Built an **end-to-end pipeline** that converts PDF documents into structured flashcard datasets by extracting, cleaning, and processing unstructured text with **LLM APIs**
- Designed a **strict JSON schema** for **AI-generated flashcards**, improving output consistency and enabling reliable parsing
- Implemented **validation logic in Python** to verify data structure, field integrity, and content quality, preventing malformed or incomplete AI responses from entering the system
- Developed an **interactive web interface** using HTML, CSS, and JavaScript to display flashcards with flip and navigation features, enabling real-time study from generated content

Wireless 4-DOF Robotic Arm | ESP32, C++, Embedded Systems

February 2026

- Designed and fabricated a **4-DOF robotic arm** using **ESP32 microcontrollers**, **servos**, and custom **3D-printed components**
- Developed **embedded C++ software** to process analog inputs and generate **PWM signals** for real-time actuator control
- Implemented **wireless control between dual ESP32 devices**, enabling synchronized motion through real-time transmission
- Engineered **modular control architecture** supporting future **inverse kinematics** and sensor integration

Investment Tracker | Java, Swing, JFreeChart

May 2025

- Built a **Java desktop app** with a **Swing-based GUI** to visualize investment growth using **side-by-side JFreeChart** graphs
- Engineered portfolio logic to calculate **asset performance** using spot prices, quantities, and historical purchase data
- Improved reliability through **exception handling** and **input validation**, ensuring consistent behavior for dynamic entries

SKILLS

Languages: C, C++, Embedded C, Python, Java, JavaScript, x86 Assembly

Web & Backend: React, HTML5, CSS, Node.js, Express.js, GraphQL

Databases & Tools: MongoDB, MySQL, SQLite, AWS S3, Git, Linux

Embedded Systems: Microcontrollers, RTOS, Hardware Interfacing, I2C, UART